Activity sheet 1.5 – Natural selection cards

Natural selection event	*Bacterial resistance to antibiotics*	*Peppered moth*
Selection Some individuals are better suited to their environment than others and so survive better.	Antibiotic-resistant bacteria are able to survive and divide (reproduce) more than non-resistant bacteria, hence passing on their antibiotic resistance.	Camouflaged moths live longer, so are able to produce more young moths with the same characteristics.
Adaptation Over many generations, the characteristics of individuals that survive better and reproduce more, become more common in the population.	More bacteria are produced, than will survive to reproduce again.	Peppered moths can be light coloured or dark coloured.
Reproduction Individuals that survive better are able to reproduce more during their lifetimes, and pass on their characteristics to the next generation.	On exposure to antibiotics, resistant bacteria survive better than non-resistant bacteria.	Peppered moths lay many eggs, but not all survive to become adult moths.
Variation There is natural variation between individuals in a population.	Some bacteria are resistant to antibiotics and some are not.	Dark moths on trees in polluted area are preyed on less than light coloured moths. Light coloured moths survive better on unpolluted trees.
Overproduction Species tend to produce more offspring than can survive to maturity.	After repeated exposure to antibiotics, antibiotic-resistance becomes more common in bacteria.	Over several generations, camouflaged moths become more common in the local population.

Thorny devil	***Great eggfly butterfly*** **(sexual selection)**	***Red-bellied black snake***
Lizards with grooves survived better because they could collect water in the dry desert. The grooves channelled morning dew from their bodies to their mouths.	A female great eggfly butterfly can lay hundreds of eggs during her lifetime, but not all eggs develop to become reproducing adults.	Because they survive better in areas with cane toads, longer snakes with smaller heads are able to reproduce more. The longer length and smaller head traits are passed on to their offspring.
Because the thorny devil lizards with grooves reproduced more, the grooves trait became more common in the population over many generations.	Because female butterflies reproduce more often with brightly coloured males, the bright wing colour becomes more common in the butterfly population.	Some red-bellied black snakes are longer than others. Head size also varies and limits the size of the prey the snake can eat.
Because the lizards with grooves survived better, they were able to reproduce more, and pass on the grooves trait to their young.	Because the brightly coloured male butterflies reproduce more, their wing colour is passed on more often to the next generation.	Snakes sometimes eat cane toads, which are toxic. Larger cane toads are more toxic than smaller ones. Snakes with larger bodies can cope with higher levels of toxin. In cane toad areas, those snakes with larger bodies and smaller heads survive the best.
Ancestral thorny devil lizards varied in body shape, with some having ridges along their body and fine grooves on their scales.	Brighter, more iridescent wings on male butterflies are more attractive to female butterflies, so these male butterflies reproduce more.	Because longer snakes with smaller heads reproduce more, this body shape becomes common in red-bellied black snake populations that live near cane toads.
Lizards lay three to ten eggs at a time, but not all of these develop into adult lizards.	Within the great eggfly butterfly species, some males have brighter iridescent blue rings on their wings than others.	Red-bellied black snakes lay up to 18 live babies at a time, but many won't survive to reproduce.